torch.compiler.substitute_in_graph#

https://pytorch.org/docs/stable/generated/torch.compiler.substitute_in_graph

Description:

Register a polyfill handler for a function, usually a C function from the C extension, to be used in place of the original function when inlining the original function in the graph. The polyfill handler is only used when inlining the original function. It is not used when the original function is called directly. In the eager mode, the decorated function calls the performant C function rather than the polyfill handler. The polyfill handler is a function that will be called in place of the original function when inlining the original function. The polyfill handler should have the same signature and the same behavior as the original function. original_fn (callable) – The original function, usually a C function, to register a polyfill handler for.

Function Signature

```
torch.compiler.substitute_in_graph(original_fn, *,
can_constant_fold_through=False, skip_signature_check=False)
[source]#

Parameters

Returns

Return type
```

Returns:

Callable[[Callable[[~_P], _R]], Callable[[~_P], _R]]

Code Examples

```
>>> import operator
>>> operator.indexOf([1, 2, 3, 4, 5], 3)
2
>>> torch.compile(operator.indexOf, fullgraph=True)([1, 2, 3, 4,
5], 3)
... # xdoctest: +SKIP("Long tracebacks")
Traceback (most recent call last):
...
torch._dynamo.exc.Unsupported: ...

>>> @torch.compiler.substitute_in_graph(operator.indexOf)
... def indexOf(a, b, /):
...     for i, item in enumerate(a):
...         if item is b or item == b:
...             return i
...     raise ValueError("sequence.index(x): x not in sequence")
>>>
>>> torch.compile(operator.indexOf, fullgraph=True)([1, 2, 3, 4,
5], 3)
2
```

Notes & Warnings

NOTE

Note The polyfill handler is only used when inlining the original function. It is not used when the original function is called directly. In the eager mode, the decorated function calls the performant C function rather than the polyfill handler.

Detailed Documentation

Documentation

Register a polyfill handler for a function, usually a C function from the C extension, to be used in place of the original function when inlining the original function in the graph.

The polyfill handler is only used when inlining the original function. It is not used when the original function is called directly. In the eager mode, the decorated function calls the performant C function rather than the polyfill handler.

The polyfill handler is a function that will be called in place of the original function when inlining the original function. The polyfill handler should have the same signature and the same behavior as the original function.

original_fn (callable) – The original function, usually a C function, to register a polyfill handler for.

can_constant_fold_through (bool, optional) – Whether the polyfill handler can be constant folded through. That is, if the polyfill handler is a pure function and its arguments are constant, the result of the polyfill handler can be constant folded during the compilation. Defaults to False.

skip_signature_check (bool, optional) – Whether to skip the signature check between the original function and the polyfill handler. Defaults to False.

A decorator that registers the polyfill handler for the original function.

Callable[[Callable[[~_P], _R]], Callable[[~_P], _R]]